Andres Arreola

Mechanical Engineering

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Education

California Polytechnic State University – San Luis Obispo (Cal Poly) June 2026

B.S. Mechanical Engineering | GPA: 3.44 | Dean's List (2x)

Relevant Coursework: Fluid Mechanics I & II, Thermodynamics I & II, Heat Transfer, Thermal System Design, Energy Conversion, Mechanical Systems Design, Measurements & Data Analysis

Involvement: Water Power Club, ASHRAE Cal Poly, Society of Hispanic Professional Engineers

Engineering Experience

Hydropower Collegiate Competition – U.S. Department of Energy Aug 2025 – Present

Design Lead | 1st Place Overall | 1st Place Build & Test | 1st Place Community Connections

- Led a cross-functional team of 4 to design a 1:5 scale cross-flow (Banki) turbine model targeting ~133 kW full-scale shaft power at 73% efficiency using hydraulic similitude laws and EES thermodynamic modeling.
- Conducted site feasibility analysis using SCADA pressure and flow data (4–31 cfs, 79–148 ft head) at a municipal pressure-reducing station to characterize annual energy potential.
- Developed experimental test procedures using a Prony brake dynamometer, turbine flow meter, and pressure transducers; analyzed performance data to generate power and efficiency curves.
- Built data reduction tools in Python (matplotlib) and Excel to produce performance curves and scale model-to-prototype power projections.
- Selected optimal turbine type via weighted decision matrix analyzing specific speed, head compatibility, and flow variability across five candidate turbine designs.

Projects

Crossflow Turbine Designer 2025

Python | crossflow-turbine-designer.streamlit.app

- Parametric design tool that takes site head and flow rate as inputs and outputs runner diameter, blade geometry, and shaft speed based on Mockmore & Merryfield and Sammartano design equations.
- Exports 3D turbine geometry compatible with SolidWorks and Blender; deployed as a live web application.

Hydropower Permit Tool 2025

Python | hydropower-permitting.up.railway.app

- Automated tool that maps site characteristics against federal and state permitting requirements for small hydropower projects.
- Identifies applicable FERC regulatory pathways, flags permitting obstacles, and generates a site feasibility summary from user-supplied inputs.

Awards & Recognition

Cal Poly Scholars 2024 – 2026

Merit program recognizing high-achieving undergraduate students from low-income backgrounds

Seal of Biliteracy – English & Spanish 2018

State-recognized achievement in English and Spanish linguistic proficiency

Technical Skills

CAD & Simulation: SolidWorks, SolidWorks FEA, SolidWorks Flow Simulation, AutoCAD

Programming & Analysis: MATLAB, Python (matplotlib, scipy), Simulink, EES, C++, Excel

Instrumentation: Pressure transducers, flow meters, thermocouples, dynamometers, data acquisition systems

Languages: English (native), Spanish (fluent)